Abstract

Trabalho de Conclusão de Curso I

Estrutura

- Contextualização
- Gap (lacuna)
- Propósito (purpose)
- Metodologia
- Resultados
- Conclusão

Reconhecimento

Abstract

Fifth generation (5G) of the cellular network is envisioned to provide diverse services to the users including high data rates, energy efficiency, and low latency. The number of Internet of Things-connected devices, multimedia applications, and mobile users are expected to grow in the next few years. The massive growth in data consumption poses new challenges such as availability and scalability in future cellular networks. Software-defined networking (SDN) is a network technology that introduces flexibility and eases the management of network architecture by separating the control plane and data plane. In this paper, we survey the relevant works that propose an SDN-based mobile core architecture in order to move toward the 5G networks and to handle the complexity of an enormous number of connected devices to the network. We propose a taxonomy in which we categorize these works based on four factors: technology adoption, control plane implementation, evaluation technique, and architecture used in these works. We also provide the shortcomings of these architectures that should be tackled in future research works.

Reconhecimento

***Abstract*—More releases of 5G operating in the millimeter wave (mmwave) range are now underway. In conjunction with the actual 5G updating process, researchers have already begun designing a new core for the 6G system. As expected, by 2030, the 6G mobile system will be employed internationally. 6G will operate at the THz gap, where Free Space Path Loss (FSPL) and atmospheric absorption loss become significant challenges. This means that new technologies need to be invented and tested to overcome these challenges and make the 6G system a reality. Among these technologies, 6G will be entirely Artificial Intelligence (AI) based. Several 6G applications are discussed in this paper, along with the general characteristics of the sub-Terahertz band and channel losses. An evaluation of the most significant 6G enabling technologies is also conducted in this paper, including AI technology, Cell-Free Massive Multiple-Input-Multi-Output (CF massive MIMO), THz band antenna technologies, Non-Orthogonal Multiple Access (NOMA), energy harvesting scenarios, Free Space Optic (FSO) backhaul technology, and 6G security technologies. The non-technological challenges issue was also considered in this paper.**

Reconhecimento

Abstract – *There are variety of methods and algorithms that can be used to overcome the ranking problem. Task ranking is one of the problems that can be solved by using a machine learning algorithm ranking problem. This work focuses on finding the right approach and corresponding algorithms in the process of ranking to be able to help people in determining which jobs have a higher priority than others. Our approach is to compare several algorithms performed in the process of ranking that are Bipartite Ranking, k-partite Ranking, and Ranking by pairwise comparison. We're used questionnaires and deployment of prototype of Intelligent Personal Assistant Agent to apply the appropriate algorithm in intelligence agent in arranging task priority in daily activity that must be done by the users. After training dataset and evaluate the validation dataset using NDCG, it is found that the collaborative ranking used have a more accurate value / lower variance test evaluation because it uses a large dataset and smaller training dataset. We found that labeling for more than 2 values it is not recommended to use a bipartite ranking if there are many repetitive data, both k-partite ranking and rank by pairwise comparison are able to be used for multi-dimensional data labeling.*

***Abstract*—Communication networks are undergoing their next evolutionary step towards 5G. The 5G networks are envisioned to provide a flexible, scalable, agile and programmable network platform over which different services with varying requirements can be deployed and managed within strict performance bounds. In order to address these challenges a paradigm shift is taking place in the technologies that drive the networks, and thus their architecture. Innovative concepts and techniques are being developed to power the next generation mobile networks. At the heart of this development lie *Network Function Virtualization* and *Software Defined Networking* technologies, which are now recognized as being two of the key technology enablers for realizing 5G networks, and which have introduced a major change in the way network services are deployed and operated. For interested readers that are new to the field of SDN and NFV this paper provides an overview of both these technologies with reference to the 5G networks. Most importantly it describes how the two technologies complement each other and how they are expected to drive the networks of near future.**

Reconhecimento

Reconhecimento

Abstract

Network slicing in 5G and beyond networks allows the network to be customized for each application or service by chaining together different virtualized network functions (VNFs) according to service requirements. The increased flexibility offered by network slicing comes at the cost of complexity in management and orchestration, which cannot be solved by traditional reactive human-in-the-loop solutions. This necessitates minimizing human intervention through the use of artificial intelligence techniques (zero touch network management). In particular, the scaling and placement of the chain of VNFs that constitute a network slice is a complex combinatorial optimization problem that is difficult to solve effectively with traditional approaches. Driven by the benefits of deep reinforcement learning (DRL) in solving various combinatorial optimization problems, in this article, we survey various DRL-based approaches to slice scaling and placement, including different ways to model the problem and benefits of various DRL techniques in addressing specific aspects of the problem. Further, we highlight key challenges and open issues in the effective use of DRL for network slice scaling and placement.